

Employee Performance

PART - A Theory

1. Define Mean, Median, Mode with an example in the context of employee salaries.

Answer

Mean:

Mean is the **average** of all salaries.

Formula:

Mean = Sum of all salaries ÷ Number of employees

Example:

- Employee salaries = **30000, 35000, 40000, 45000, 50000**

$$\begin{aligned}\text{Mean} &= (30000 + 35000 + 40000 + 45000 + 50000) \div 5 \\ &= \mathbf{40000}\end{aligned}$$

Real-life Example:

A company wants to know the **average salary** of employees.

Python Code

```
import numpy as np
```

```
salary = [30000, 35000, 40000, 45000, 50000]
```

```
print("Mean Salary:", np.mean(salary))
```

Median

Definition:

Median is the **middle value** after arranging the salaries in order.

Example:

- 30000, 35000, **40000**, 45000, 50000

Median = **40000**

Real-life Example:

Median is useful when a few employees have very high salaries.

Python Code

```
import numpy as np
```

```
salary = [30000, 35000, 40000, 45000, 50000]
```

```
print("Median Salary:", np.median(salary))
```

Mode

Definition:

Mode is the salary that appears **most frequently**.

Example:

30000, 35000, 35000, 40000, 50000

Mode = **35000**

Real-life Example:

Most employees in a company earn ₹35,000.

Python Code

```
from statistics import mode

salary = [30000, 35000, 35000, 40000, 50000]

print("Mode Salary:", mode(salary))
```

2. Differentiate between Range and Variance with examples.

Range	Variance
Difference between highest and lowest value.	Measures how much data is spread from the average.
Easy to calculate.	More detailed measurement.

Example

Salaries:

30000, 35000, 40000, 45000, 50000

Range = 50000 – 30000 = **20000**

Variance shows how far each salary is from the average.

Real-life Example:

A company checks whether employee salaries are very different or almost the same.

Python Code

```
import numpy as np
```

```
salary = [30000, 35000, 40000, 45000, 50000]
```

```
print("Range:", max(salary) - min(salary))
```

```
print("Variance:", np.var(salary))
```

3. What is the difference between Normal Distribution and Poisson Distribution?

Normal Distribution

Continuous data

Bell-shaped curve

Used for height, salary, marks

Poisson Distribution

Count data

Counts events

Used for number of calls, accidents

Real-life Example

Normal Distribution:

Employee salaries in a company.

Poisson Distribution:

Number of customer complaints received in one day.

Python Code

```
from scipy.stats import norm, poisson
```

```
print(norm.pdf(50, 50, 10))
```

```
print(poisson.pmf(3, 5))
```

4. Explain Skewness and give one workplace-related example.

Answer

Skewness tells whether data is **balanced or not**.

- Positive Skew → Long tail on the right
- Negative Skew → Long tail on the left
- Zero Skew → Perfectly balanced

Real-life Example

Most employees earn ₹35,000, but a few managers earn ₹2,00,000.

The salary data becomes **positively skewed**.

Python Code

```
from scipy.stats import skew
```

```
salary = [30000,35000,35000,36000,37000,200000]
```

```
print("Skewness:", skew(salary))
```

5. Define Conditional Probability and explain how it applies in promotions.

Answer

Conditional Probability means the probability of an event **when another event has already happened**.

Formula

$$P(A|B) = P(A \text{ and } B) / P(B)$$

Real-life Example

Find the probability that an employee gets promoted **if** their performance score is above 80.

Python Code

```
promoted = 40
```

```
high_performance = 50
```

```
probability = promoted / high_performance
```

```
print("Conditional Probability:", probability)
```

6. Explain Independent vs Mutually Exclusive Events with examples.

Independent Events

Two events that **do not affect each other**.

Example

Employee A gets promoted.

Employee B gets promoted.

These events are independent.

Mutually Exclusive Events

Two events **cannot happen together**.

Example

An employee can be either

- Promoted
- Not Promoted

Both cannot happen at the same time.

Python Code

```
promotion = "Yes"
```

```
not_promotion = "No"
```

```
print("Employee Promotion:", promotion)
```

```
print("Employee Not Promotion:", not_promotion)
```

7. What does Bayes Theorem help in real-world decision-making?

Answer

Bayes Theorem helps us **update the probability** of an event when new information is available.

Real-life Example

A company predicts whether an employee will be promoted based on their performance score.

Python Code

$P_A = 0.6$

$P_{B_given_A} = 0.8$

$P_B = 0.7$

$bayes = (P_{B_given_A} * P_A) / P_B$

`print("Bayes Probability:", bayes)`

8. Write a short note on PCA (Principal Component Analysis) in simple words.

Answer

PCA (Principal Component Analysis) is a technique used to **reduce the number of features** in a dataset while keeping the most important information.

It makes the data **smaller, faster, and easier to analyze**.

Real-life Example

An employee dataset has **20 columns** like Age, Salary, Experience, Projects, Working Hours, etc.

PCA reduces them to **2 or 3 important features** for easier analysis and visualization.

Python Code

```
from sklearn.decomposition import PCA
```

```
import numpy as np
```

```
X = np.array([
```

```
    [25,30000],
```

```
    [30,40000],
```

```
    [35,50000]
```

```
])
```

```
pca = PCA(n_components=1)
```

```
result = pca.fit_transform(X)
```

```
print(result)
```